

Deep reinforcement learning-based attitude control for spacecraft using control moment gyros

Snyoll Oghim^a, Junwoo Park^a, Hyochoong Bang^{a,*}, Henzeh Leeghim^b

^a Korea Advanced Institute of Science and Technology, Yuseong-gu, Daejeon 34141, Republic of Korea

^b Chosun University, Dong-gu, Gwangju 61452, Republic of Korea

Received 11 March 2024; received in revised form 16 June 2024; accepted 29 July 2024

Available online 7 August 2024

Abstract

This paper addresses the development of an attitude control system that steers control moment gyros (CMGs) based on deep reinforcement learning (DRL) for agile spacecraft. The proposed DRL-based attitude control system learns CMG steering strategies to achieve the desired attitude, thus potentially bypassing the singularity issues inherent in the CMG cluster. In particular, it is designed in two-phases to apply the DRL technique efficiently. In the first phase, the attitude control is performed based on DRL up to a certain tolerance, after which it switches to conventional control and steering law for stabilization in the second phase. The rapid pointing capability of the proposed DRL-based attitude control system is demonstrated for an agile spacecraft equipped with pyramid-type single gimbal control moment gyros. Additionally, in realistic scenarios of pointing multiple targets on the ground, the momentum vector recovery that the CMG system needs to consider is also briefly discussed.

© 2024 COSPAR. Published by Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Keywords: Spacecraft; Attitude control; Control moment gyros; Reinforcement learning

1. Introduction

Attitude control is necessary for spacecraft operations, especially when large-angle slew or fast maneuvers are required; the implementation plays an important role considering future space missions using spacecraft such as military surveillance and reconnaissance or deep space explorations. Over the decades, many researchers have studied spacecraft attitude control, and various controllers considering different types of actuators have been developed Wang and Shtessel (1998), Yeh (2010), Petersen et al. (2017), Bunryo et al. (2021).

Changing the orientation of spacecraft can be classified into passive and active methods. Passive methods utilize air drag, earth gravity, or earth's magnetic field, so the torque generated is small, and thus is often used in micro-satellites Sutherland et al. (2018). The active methods perform attitude control by generating torque from thrusters, magnetic torquers, and reaction devices. These internal reaction devices include reaction wheels (RWs) or control moment gyros (CMGs) for reorientation maneuvers.

A Single-Gimbal Control Moment Gyro (SGCMG) serves as a momentum exchange mechanism in spacecraft and is employed for attitude control. This system generates gyroscopic torque by rotating a flywheel mounted within a gimbal. SGCMGs, typically in four clusters, provide redundancy and enable comprehensive 3-axis attitude control of the spacecraft. Nonetheless, SGCMGs have challenges due to geometric singularities, in which generating

* Corresponding author.

E-mail addresses: s.oghim@kaist.ac.kr (S. Oghim), junwoopark@kaist.ac.kr (J. Park), hcbang@kaist.ac.kr (H. Bang), h.leeghim@controla.re.kr (H. Leeghim).

torque in specific directions becomes unfeasible. Such singularity issues arise when the momentum vectors of the SGCMGs align within the same plane or converge along a singular line. The singularity problem associated with the CMG system becomes a significant limitation on the maneuverability of agile spacecraft [Leeghim et al. \(2018\)](#). Over the past decades, many strategies have been explored to avoid/escape the singularity problem, aiming to enhance spacecraft attitude control systems' operational efficiency and reliability.

The most basic steering law is the pseudo-inverse (PI) steering law, which provides a minimal two-norm solution but does not explicitly avoid singularity [Davis \(1972\)](#). A typical approach to avoid singularities is to add null motion using gradient vectors to the pseudo-inverse solution [Cornick \(1979\)](#). These methods work well for certain SGCMGs but are also considered insufficient for pyramid-type SGCMGs.

In addition to the null motion approaches, various forms of singularity-robust (SR) inverse steering laws have been employed. These methods attempt to escape singularities by allowing a torque error [Bedrossian et al. \(1990\)](#), [Wie et al. \(2000\)](#), [Wie et al. \(2001\)](#). Although the SR method is relatively straightforward, only a specific class of singularity can be remedied; it is robust to a particular situation, yet not universal. For instance, an off-diagonal singularity robustness steering law is proposed to handle a specific class of saturation singularity by introducing a time-dependent modulation function into the gimbal rate construction [Wie \(2005\)](#).

To improve singularity avoidance/escape further, another study proposed a global singularity avoidance steering law that employs a switching control strategy to avoid all types of singularities [Paradiso \(1992\)](#). However, this method requires high computational cost and is unsuitable for real-time applications. In order to relieve the computational burden, research has been conducted to attempt to geometrically avoid singularities introducing a surface cost function [Takada et al. \(2010\)](#). In addition, various other steering methods are being studied, such as steering law that considers the subsequent step [Leeghim et al. \(2009\)](#), steering law using artificial potential functions [Seo et al. \(2019\)](#), and steering law that applies game theory [Lee et al. \(2005\)](#), [Hua et al. \(2023\)](#). Research has also been conducted using Rapidly-exploring Random Tree (RRT), a representative algorithm used for path planning [Mony and Paranjape \(2022\)](#).

Recently, studies on ways to utilize machine learning technique have been attracting attention. Research has been conducted on adaptive control using basic neural networks [Leeghim and Kim \(2015\)](#), and there is research using machine learning techniques such as deep neural network and random forest classifier to predict for trajectory to predict the required null motion [Papakonstantinou et al. \(2022\)](#). Although it is not a system consisting of several CMG, in another case, an adaptive radial bias function neural network is utilized to control the underactuated

CMG [Montoya-Cháirez et al. \(2021\)](#). Aside from this, much research is in progress to utilize machine learning in CMG systems, but most of the research focuses on fault diagnosis [Zhao et al. \(2022\)](#), [Yuandong et al. \(2022\)](#), [Zhang et al. \(2023\)](#), and in particular, cases of utilizing reinforcement learning are still insufficient.

In this paper, we apply deep reinforcement learning (DRL) techniques for attitude control of agile spacecraft. These techniques are utilized to learn the steering strategy of SGCMGs for achieving the desired attitude. By directly steering the SGCMGs without relying on their mathematical models, we present the potential to circumvent the singularity problem associated with the CMG system effectively. However, executing the entire spacecraft maneuver solely with DRL techniques is inefficient. This inefficiency becomes more pronounced during the attitude stabilization phase, where the goal is to achieve the desired attitude and maintain the spacecraft's angular rates at zero. Consequently, the DRL-based attitude control system proposed in this paper is divided into two phases. In the first phase, DRL techniques are employed for large-angle maneuvers. The system transitions to the second phase once the attitude error is reduced to within a specific tolerance. In this phase, conventional control and steering law are applied to achieve precise attitude control and stabilization.

The anticipated advantages of this study are as follows. First, it enables the full utilization of the CMG system's performance without the singularity problem, offering a novel perspective that can surmount the limitations of existing steering laws. Second, it can be deployed without additional training, even in the presence of model uncertainties, disturbances, noise, or changes in the CMG system's configuration. This underscores the superior versatility of the proposed approach compared to existing machine learning-based steering laws with specific application ranges. Third, since reinforcement learning learns through interaction with the environment, the need to create a separate training dataset is obviated. This not only saves time and effort in securing training data but also enhances the practicality and efficiency of our approach.

The primary objective of this paper is to present the validity and advantages of applying DRL techniques to the attitude control system of agile spacecraft equipped with SGCMGs. To achieve this, we first provide a concise overview of the spacecraft attitude control problem alongside an introduction to DRL techniques. Subsequently, we formulate the problem within the context of the DRL framework. Lastly, we investigate our proposed attitude control system through detailed simulation results, demonstrating its effectiveness in agile spacecraft.

2. Attitude control using SGCMGs

This paper addresses the spacecraft attitude control problem using SGCMGs. This section briefly introduces the fundamental principles for spacecraft attitude dynamics

and kinematics. First, the spacecraft attitude is expressed using the quaternion kinematic differential equations.

$$\dot{\mathbf{q}} = \frac{1}{2} \mathbf{q} \otimes \Omega \quad (1)$$

where the unit quaternion convention is utilized, i.e., $\mathbf{q} = [q_w, \mathbf{q}_v^T]^T$, $\otimes$ is used to represent quaternion multiplication, $\Omega = [0, \boldsymbol{\omega}^T]^T$, and $\boldsymbol{\omega} = [\omega_x, \omega_y, \omega_z]^T$ are the angular velocity vectors of the spacecraft. Moreover, q_w and $\mathbf{q}_v^T$ denote the scalar and vector components of the quaternions, respectively.

If the desired quaternions are given $\mathbf{q}_d$, one can obtain the attitude error quaternions with the conjugate of the desired quaternions $\mathbf{q}_d^*$ as follows,

$$\mathbf{q}_e = [q_{we}, \mathbf{q}_{ve}^T]^T = \mathbf{q}_d^* \otimes \mathbf{q} \quad (2)$$

Then, the attitude error quaternions can be defined such that

$$\dot{\mathbf{q}}_e = \frac{1}{2} \mathbf{q}_e \otimes \Omega \quad (3)$$

Next, consider Euler's rotational equations of motion for a rigid spacecraft described by

$$\dot{H}_{sc} + \boldsymbol{\omega} \times H_{sc} = \boldsymbol{\tau}_{ext} \quad (4)$$

where H_{sc} is the total angular momentum vector of the spacecraft, and $\boldsymbol{\tau}_{ext}$ is the external torque. The total angular momentum of the system consists of the angular momentum of the spacecraft main body and the angular momentum contributed by SGCMGs.

$$H_{sc} = J\boldsymbol{\omega} + H_{cmg} \quad (5)$$

where J is the moment of inertia matrix of the spacecraft, and H_{cmg} is the angular momentum vector of SGCMGs. Substituting Eq. 5 into Eq. 4 gives,

$$J\dot{\boldsymbol{\omega}} + \dot{H}_{cmg} + \boldsymbol{\omega} \times (J\boldsymbol{\omega} + H_{cmg}) = \boldsymbol{\tau}_{ext} \quad (6)$$

The control torque generated by the SGCMGs is defined as

$$\boldsymbol{\tau} = -\dot{H}_{cmg} - \boldsymbol{\omega} \times H_{cmg} \quad (7)$$

Then substitution of Eq. 6 into Eq. 5 yields

$$J\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times J\boldsymbol{\omega} = \boldsymbol{\tau} \quad (8)$$

where the external torque is assumed to be zero.

Depending on how SGCMGs are arranged, SGCMG clusters exhibit slightly different characteristics. The SGCMG cluster considered in this paper is a pyramid-type cluster, as shown in Fig. 1, with an almost uniform momentum magnitude in all directions. The total angular momentum of SGCMGs, each of which is installed having a skew-angle β and angular momentum h such that

$$H_{cmg} = h \begin{bmatrix} -\cos \beta \sin \delta_1 - \cos \delta_2 + \cos \beta \sin \delta_3 + \cos \delta_4 \\ \cos \delta_1 - \cos \beta \sin \delta_2 - \cos \delta_3 + \cos \beta \sin \delta_4 \\ \sin \beta \sin \delta_1 + \sin \beta \sin \delta_2 + \sin \beta \sin \delta_3 + \sin \beta \sin \delta_4 \end{bmatrix} \quad (9)$$

Then, the time derivative of H_{cmg} leads to

$$\dot{H}_{cmg} = A(\boldsymbol{\delta})\dot{\boldsymbol{\delta}} \quad (10)$$

where $\boldsymbol{\delta} = [\delta_1, \delta_2, \delta_3, \delta_4]^T$ is the gimbal angle vector and the corresponding Jacobian matrix is given by,

$$A(\boldsymbol{\delta}) = h \begin{bmatrix} -\cos \beta \cos \delta_1 & \sin \delta_2 & \cos \beta \cos \delta_3 & -\sin \delta_4 \\ -\sin \delta_1 & -\cos \beta \cos \delta_2 & \sin \delta_3 & \cos \beta \cos \delta_4 \\ \sin \beta \cos \delta_1 & \sin \beta \cos \delta_2 & \sin \beta \cos \delta_3 & \sin \beta \cos \delta_4 \end{bmatrix} \quad (11)$$

If the desired torque $\boldsymbol{\tau}_d$ is given, the gimbal rates required can be calculated as follows.

$$\dot{\boldsymbol{\delta}} = A^T (AA^T)^{-1} \boldsymbol{\tau}_d \quad (12)$$

This equation is called the pseudo-inverse steering law. However, the existence of the pseudo-inverse of the Jacobian matrix cannot be guaranteed since the elements of the Jacobian matrix vary with the gimbal angle. If the angular momentum vectors of the SGCMGs lie in the same plane, the Jacobian matrix is not full rank, and the desired gimbal rates cannot be obtained. This is known as the mathematical singularity of SGCMGs, which has been addressed in the introduction. The singularity problem has already been investigated in many previous studies [Wie \(2004\)](#), [Yoon and Tsotras \(2004\)](#), [Liang and Xu \(2005\)](#). In this paper, we propose a new approach to avoid this singularity issue using the RL technique.

3. Reinforcement learning

Deep reinforcement learning focuses on making a series of optimal decisions to maximize the expected reward. Especially for a given Markov decision process (MDP), the DRL approach is a mathematical framework that can infer the optimal choice of actions, even under a stochastic environment [François-Lavet et al. \(2018\)](#).

In an MDP, an agent interacts with an environment over all discrete time steps. At each time step t , the agent observes the current state, s_t , and takes an action to take, a_t . The environment then transitions to a new state, s_{t+1} , and the agent receives a reward, r_t , based on the transition and the action taken.

$$r_t = \mathcal{R}(s_t, a_t, s_{t+1}) \quad (13)$$

where $\mathcal{R}$ denotes a reward function. This cyclic process is repeated until one of the user-defined termination or truncation conditions is met, such as exceeding the maximum number of steps. One cycle is called an episode. The agent's goal is to maximize the cumulative reward over whole steps, also known as the return.

$$R_t = \sum_{i=t}^T \gamma^{i-t} r_i \quad (14)$$

where T is the step at the end of an episode, and $\gamma \in (0, 1]$ is a discount factor, which denotes an attenuation of the reward based on how far off it can be obtained in the

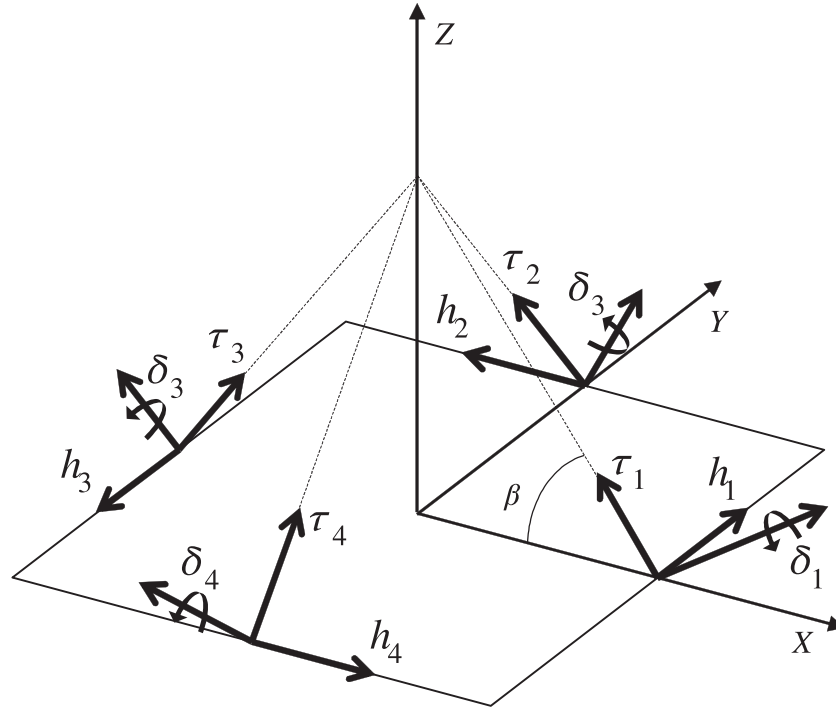


Fig. 1. Pyramid-type of four single-gimbal CMGs.

future. In other words, the objective in DRL is to find a policy, π , that maximizes the expected return when the agent acts based on it. Therefore, DRL can be seen as an optimization problem expressed as follows:

$$\mathcal{J}(\pi) = \mathbb{E}_{\pi}[R_t] \quad (15)$$

$$\pi^* = \arg \max_{\pi} \mathcal{J}(\pi) \quad (16)$$

where π^* means the optimal policy.

DRL algorithms for MDP employ a value function to make a decision. The value functions are generally categorized into two branches: a state-value function and an action-value function. The state-value function is used to estimate the expected return of being in a particular state while following a given policy. The action-value function, the Q-function, estimates the expected return of taking a particular action in a given state and following a given policy. Each is defined as follows:

$$V^{\pi}(s_t) = \mathbb{E}_{\pi}[R_t | s_t] \quad (17)$$

$$Q^{\pi}(s_t, a_t) = \mathbb{E}_{\pi}[R_t | s_t, a_t] \quad (18)$$

Accurately calculating a value function in DRL is a challenging task. Therefore, it is a common practice to estimate the value function. The value function approximation is typically carried out via sampling data through interactions between the agent and the environment. To estimate the value function, we commonly use the Bellman equations [Sutton and Barto \(2018\)](#). Bellman equations present the relationship between the current and subsequent state-value function (or action-value function). The current value function can be updated through the Bellman equations with the sampled data. The Bellman equations for the value functions are described as follows:

$$V^{\pi}(s_t) = \mathbb{E}_{\pi}[r_t + \gamma V^{\pi}(s_{t+1})] \quad (19)$$

$$Q^{\pi}(s_t, a_t) = \mathbb{E}_{\pi}[r_t + \gamma \mathbb{E}_{\pi}[Q^{\pi}(s_{t+1}, a_{t+1})]] \quad (20)$$

This method can be employed with model-based and model-free algorithms, the two broad categories of DRL algorithms. Model-based algorithms assume some prior knowledge of the environment and its behavior, whereas model-free algorithms learn through trial-and-error interactions with the environment. This paper uses a model-free DRL approach because the mathematical model of the SGCMGs cluster has a singularity problem.

The model-free approach consists of two principles for updating the policy: Policy optimization and Q-learning. Q-learning is used to learn the optimal action-value function for a given environment. This algorithm estimates the expected Q-value in Eq. 20 and updates it using the Bellman equations. Many different algorithms, such as Deep Q-Network [Mnih et al. \(2013\)](#), update the policy in an off-policy way, and most of them are known to be sample-efficient.

On the other hand, policy optimization is a technique used to learn a policy function that maps states to actions. Unlike Q-learning, which estimates the optimal action-value function, policy optimization directly learns the policy that maximizes the expected reward. There are various algorithms for policy optimization, such as Trust Region Policy Optimization (TRPO) [Schulman et al. \(2015\)](#) and Proximal Policy Optimization (PPO) [Schulman et al. \(2017\)](#). Those algorithms are subject to lower sample efficiency due to using an on-policy way to update the policy. Yet, they are easy to implement and exhibit a stable learning curve.

Lastly, some algorithms are a compromise between Q-learning and policy optimization, taking the strengths and weaknesses of both techniques. These include Deep Deterministic Policy Gradient (DDPG) [Lillicrap et al. \(2015\)](#), Twin Delayed Deep Deterministic Policy Gradient (TD3) [Fujimoto et al. \(2018\)](#), and Soft Actor-Critic (SAC) algorithms [Haarnoja et al. \(2018\)](#).

3.1. Soft actor-critic

There are two main challenges to consider when applying the DRL technique to the attitude control problem using SGCMGs. First, since the desired attitude is achieved through gimbal rates, the action, a_t , and observation, s_t , should be defined in continuous space. Second, as described in the previous section, the mathematical model of SGCMGs has a singularity problem. Therefore, inferring from the sampling data without explicitly considering the model is necessary, which requires high sample efficiency. To tackle these challenges, this paper employs the soft actor-critic (SAC) algorithm.

SAC is a model-free approach that is used for continuous control tasks. It is based on the maximum entropy DRL framework, which aims to maximize both the expected reward and the entropy of the policy. In the SAC algorithm, entropy denotes a quantity that measures the randomness of the policy. Increasing entropy implies more exploration that can be interpreted as preventing the policy from converging to bad local optimum [Haarnoja et al. \(2018\)](#). The entropy of a random variable x with the probability distribution function P is defined as

$$\mathcal{H}(P) = \mathbb{E}_P[-\log P(x)] \quad (21)$$

Adding the entropy term to Eq. 15, one can define the new objective function as [Ziebart \(2010\)](#),

$$\mathcal{J}(\pi) = \mathbb{E}_\pi \left[\sum_{t=0}^T \gamma^t (r_t + \alpha \mathcal{H}(\pi(\cdot|s_t))) \right] \quad (22)$$

where $\alpha > 0$ is a temperature parameter that controls the trade-off between expected reward and entropy, and $\mathcal{H}(\pi(\cdot|s_t))$ is the entropy of the policy.

As shown in [Algorithm 1](#), SAC involves five neural network models: one actor network, two critic networks, and two target critic networks. The actor network generates actions based on the policy, while the critic networks estimate the state-action values, representing the expected rewards for a given state and action. The critic networks are trained using the error loss between the estimated state-action values and the actual rewards the agent receives.

One key feature of SAC incorporates a replay buffer, which stores transitions from the environment for later training. Using data from the replay buffer, SAC can learn from a broader range of experiences, leading to more robust policies. The replay buffer also allows for decoupling the agent's exploration from the learning process, improving training stability.

Overall, SAC is a practical algorithm for learning policies in continuous action spaces. Its use of a stochastic policy and replay buffer makes it particularly useful for exploration and robustness. Additionally, its off-policy nature and use of two critic networks help to prevent overestimation and accelerate training.

Algorithm 1. Soft Actor-Critic

Algorithm 1: Soft Actor-Critic

```

Initialize policy parameter  $\theta$ , Q-function parameters  $\phi_1, \phi_2$ , empty replay buffer  $\mathcal{D}$ 
Set target parameters equal to main parameters  $\phi_{trg,1} \leftarrow \phi_1, \phi_{trg,2} \leftarrow \phi_2$ 
for each iteration do
    Observe state  $s_t$  and select action  $a_t \sim \pi_\theta(\cdot|s_t)$ 
    Execute  $a_t$  in the environment and observe  $s_{t+1}, r_t$ , and done signal,  $d$ , indicating the agent reaches terminal condition
    Store  $(s_t, a_t, r_t, s_{t+1}, d)$  in replay buffer  $\mathcal{D}$ 
    If  $d$  is TRUE, reset environment and agent
    if It's time to update then
        for  $j = 1, 2, \dots, N_{updates}$  do
            Randomly sample a batch of transitions,  $\mathcal{B} = \{(s_t, a_t, r_t, s_{t+1}, d)\}$  from  $\mathcal{D}$ 
            Compute targets for Q-functions
             $y(r_t, s_{t+1}, d) = r_t + \gamma(1-d) \left( \min_{i=1,2} Q_{\phi_{trg,i}}(s_{t+1}, a_{t+1}) - \alpha \log \pi(\tilde{a}_{t+1}|s_{t+1}) \right)$ ,  $\tilde{a}_{t+1} \sim \pi_\theta(\cdot|s_{t+1})$ 
            Update Q-functions by one step of gradient descent using,
             $\nabla_{\phi_i} \frac{1}{|\mathcal{B}|} \sum_{(s_t, a_t, r_t, s_{t+1}, d) \in \mathcal{B}} [Q_{\phi_i}(s_t, a_t) - y(r_t, s_{t+1}, d)]^2$  for  $i = 1, 2$ 
            Update policy by one step of gradient ascent using,
             $\nabla_\theta \frac{1}{|\mathcal{B}|} \sum_{s_t \in \mathcal{B}} \left[ \min_{k=1,2} Q_{\phi_k}(s_t, \tilde{a}_\theta(s_t)) - \alpha \log \pi_\theta(\tilde{a}_\theta(s_t)|s_t) \right]$ 
            where  $\tilde{a}_\theta$  is a sample from  $\pi(\cdot|s_t)$  which is differentiable with respect to  $\theta$ 
            Update target network with
             $\phi_{trg,k} \leftarrow \rho \phi_{trg,k} + (1-\rho)\phi_k$  for  $k = 1, 2$ 
        end
    end
end

```

4. Deep reinforcement learning-based attitude control

This section introduces the approach to addressing the attitude control problem of spacecraft equipped with SGCMGs within the DRL framework, utilizing the SAC algorithm. The problem formulation involves defining a suitable MDP and an appropriate reward function. The environment in which the agent interacts is characterized by the rotational equations of motion and quaternion kinematics presented in the previous section. Within this environment, the observations available to the agent are defined as follows:

$$s_t = (\mathbf{q}_e, \boldsymbol{\omega}, \boldsymbol{\delta}) \quad (23)$$

Here, the reason for using the attitude error quaternions is for the consistency of the goals the agent should achieve. No matter the desired attitude, the state the agent must reach is always $\mathbf{q}_e = [1, 0, 0, 0]^T$. Attitude error quaternions are also helpful in the design of reward functions.

To achieve the desired attitude of a spacecraft equipped with SGCMGs, it is necessary to control the gimbal rates of the SGCMGs to generate the gyroscopic torque. Conventional steering laws utilize the pseudo-inverse of SGCMGs, as shown in Eq. 12, to determine the gimbal rates according to the desired torque derived from the control law. However, the availability of a pseudo-inverse is not always guaranteed. Therefore, we propose using DRL techniques to derive gimbal rates directly from sampled data without calculating the pseudo-inverse. For this purpose, we define the agent's actions as gimbal rates $\dot{\boldsymbol{\delta}}$, and the action space is also defined using the maximum gimbal rates $\dot{\boldsymbol{\delta}}_{max}$.

Designing an appropriate reward function is crucial for formulating a complete MDP for the attitude control problem, considering the environment, states, and actions. The reward function forms the foundation of agent training and significantly influences the policy that dictates the principles of action within the DRL framework. It is considered one of the most critical components in the DRL framework. Nonetheless, there are no definitive guidelines for designing reward functions. The design of the reward function varies depending on the problem being addressed and often requires a considerable amount of trial and error.

The reward function proposed in this paper is designed with three key factors. The first is the spacecraft's agility to achieve the desired attitude, for which we directly utilized attitude error as a criterion. The second involves minimizing the spacecraft's angular rates once it has achieved the desired attitude, directly utilizing the spacecraft's angular rates to accomplish this. The third and final element ensures that the gimbal angle returns to its initial setting when the desired attitude is successfully achieved. This consideration is crucial for operational aspects of the CMG system, especially in real scenarios pointing to multiple targets. Additionally, there is a potential benefit in mitigating the risk of falling into singularity issues with SGCMGs

when switching to conventional control and steering for precise attitude control and stabilization.

$$r_t = r_1 + r_2 \quad (24)$$

Each term is defined such that

$$r_1 = -k_q \|\mathbf{q}_{ve}\|, \quad r_2 = \begin{cases} -k_1 \log(\|\boldsymbol{\omega}\| + \|\boldsymbol{\delta}_{err}\|) + k_2, & \text{if } \|\mathbf{q}_{ve}\| \leq b_q \\ 0, & \text{otherwise} \end{cases}$$

where k_q, k_1 , and k_2 are positive constants to shape the reward function.

In Eq. 24, the first terms represent the attitude error quaternions. The second term is related to the angular rates and the gimbal angles error, $\boldsymbol{\delta}_{err} = \boldsymbol{\delta}_0 - \boldsymbol{\delta}_t$, where $\boldsymbol{\delta}_0$ is the initial gimbal angles of the SGCMGs and $\boldsymbol{\delta}_t$ is its the gimbal angles at time step t . Using the logarithm reflects a more significant reward as the angular rates and the angle error decreases. This reward term is only applied when the spacecraft's attitude error satisfies a specific tolerance, b_q , which is also the parameter for switching to the second phase of the proposed DRL-based attitude control system. In other words, decreasing b_q enhances the accuracy of control but increases the learning time.

The coefficients used in the reward function are summarized in Table 1. k_1 and k_2 are set to ensure that, upon achieving the goal, accumulated negative reward value over the episode is converted into positive value, making it clear to the agent that the goal has been achieved. b_q is set to 0.04, a value determined through Monte-Carlo simulations to be the level at which the conventional control law and steering law used in the second phase of the proposed attitude control system can avoid singularity issues.

The process of the agent interacting with the environment and progressing through an episode can be summarized as follows: At each step, the agent takes an action defined by the gimbal speed of SGCMGs, as derived from its policy. The environment calculates the spacecraft's state for the next step based on the torque generated from the agent's action. The agent repeats this process until the episode termination or truncation conditions are met. The entire process is illustrated in Fig. 2.

4.1. Training scenarios

This section introduces the considerations for training scenario. We assume that the spacecraft performs a rest-to-rest (R2R) maneuver. This means that the spacecraft's angular momentum must be zero both before starting and after achieving the desired attitude control. Training begins with initializing the agent and the environment, during which a desired attitude is randomly selected, and the

Table 1
Hyperparameters used to train the agent.

k_q	k_1	k_2	b_q
1.0	100	500	0.04

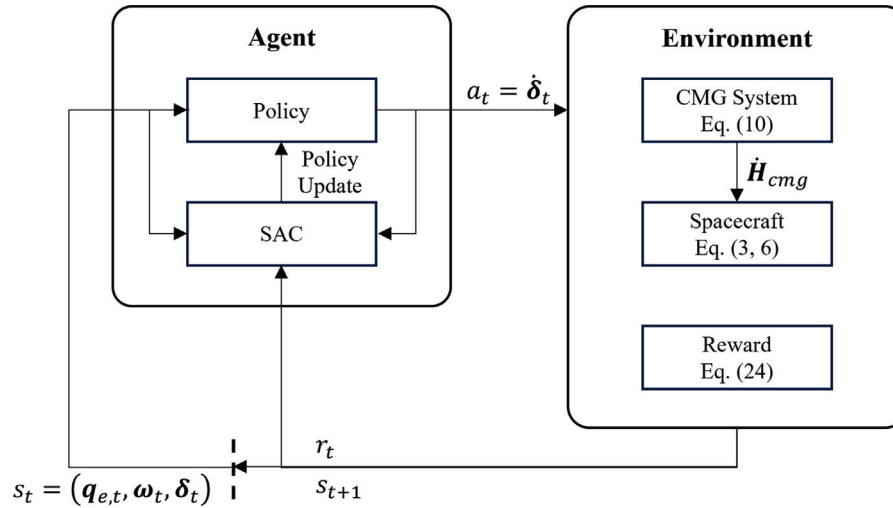


Fig. 2. A representation of the DRL single-step process for the CMG system-based attitude control problem.

attitude error quaternion is calculated based on this selection. The spacecraft's angular rates is initialized to zero, and the initial gimbal angles of the SGCMGs are set to a user-selected configuration that ensures the total angular momentum of the SGCMGs is zero.

We also consider a maximum time of 30 s for a maneuver and a time step of 0.1 s. Therefore, the maximum number of steps per episode is 300. This maximum step serves as the truncation condition for an episode; if the agent achieves or fails to achieve the set tolerance within 300 steps, the episode terminates. When an episode ends, the state and reward information constituting that episode are stored in the replay buffer. The episodes stored in the replay buffer are used to update the agent's policy. By repeating this process, the agent learns how to steer the SGCMGs to achieve the desired attitude for the spacecraft. The detailed configurations of the spacecraft and SGCMGs used in training are summarized in Table 2, with initial gimbal angles referenced from literature Kuhn and Rodriguez (1995).

There is a concern when using DRL for spacecraft attitude control, especially when trying to maintain a target attitude by reducing the spacecraft's angular rates to zero. The issue arises when the agent opts for abrupt maneuvers instead of halting to maintain the attitude. Although this problem can be mitigated by incorporating an energy-

minimizing term into the reward function, it adversely affects the spacecraft's maneuverability and extends the training duration.

The proposed DRL-based attitude control system distinguishes precision attitude control and stabilization phases from an overall attitude control scheme to solve that consideration and efficiently utilize the DRL technique. The first phase involves controlling the attitude for large-angle maneuvers through DRL. Once the attitude error has reached a specific tolerance, the system transitions to the second phase, where conventional attitude control and steering law are employed for precise attitude control and stabilization.

The conventional attitude control law used in the second phase is referenced in Wie et al. (2001).

$$\tau_c = -J \left\{ 2k \text{sat}_{L_i} \left(\mathbf{q}_e + \frac{1}{T} \int \mathbf{q}_e \right) + c\boldsymbol{\omega} \right\} \quad (25)$$

The variable limiter and the saturation function is defined as

$$L_i = \left(\frac{c}{2k} \right) \min \left\{ \sqrt{4a_i |q_{e,i}|}, |\omega_i|_{\max} \right\}$$

$$\text{sat}_L(e) = \begin{cases} L & \text{if } e \geq L \\ e & \text{if } |e| < L \\ -L & \text{if } e \leq -L \end{cases}$$

where k and c are controller gain to be properly determined.

The spacecraft must maintain consistent maneuverability throughout the entire mission duration. Since the gimbal angle positions of the CMG system are closely related to the spacecraft's rotational maneuverability, to keep the spacecraft's maneuverability consistent, the gimbal angles of the CMG system must always be the same at the start and end of every maneuver. In the first phase of the proposed attitude control system, the reward function is designed to ensure that the gimbal angles of the SGCMGs

Table 2
Specification of Spacecraft equipped with SGCMGs used to train the agent Wie et al. (2002).

Parameter	Value	Unit
Moment of inertia, J	$\text{diag}(21400, 20100, 5500)$	kg-m ²
Initial quaternions, $\mathbf{q}$	$[1, 0, 0, 0]^T$	-
Initial angular rates, $\boldsymbol{\omega}$	$[0, 0, 0]^T$	deg/s
Skew-angle, β	53.13	deg
Angular momentum of flywheel, h	1000	Nm
Initial gimbal angle, δ_0	$[45, -45, 45, -45]^T$	deg
Maximum gimbal rate, $\dot{\delta}_{\max}$	1.0	rad/s

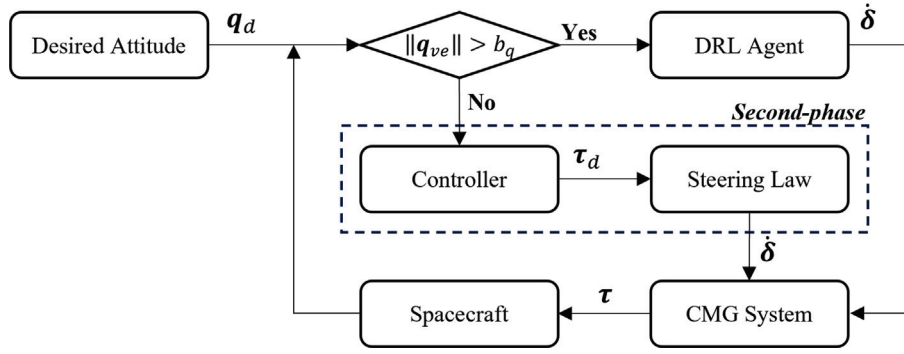


Fig. 3. Conceptual flowchart of the DRL-based Attitude Control System.

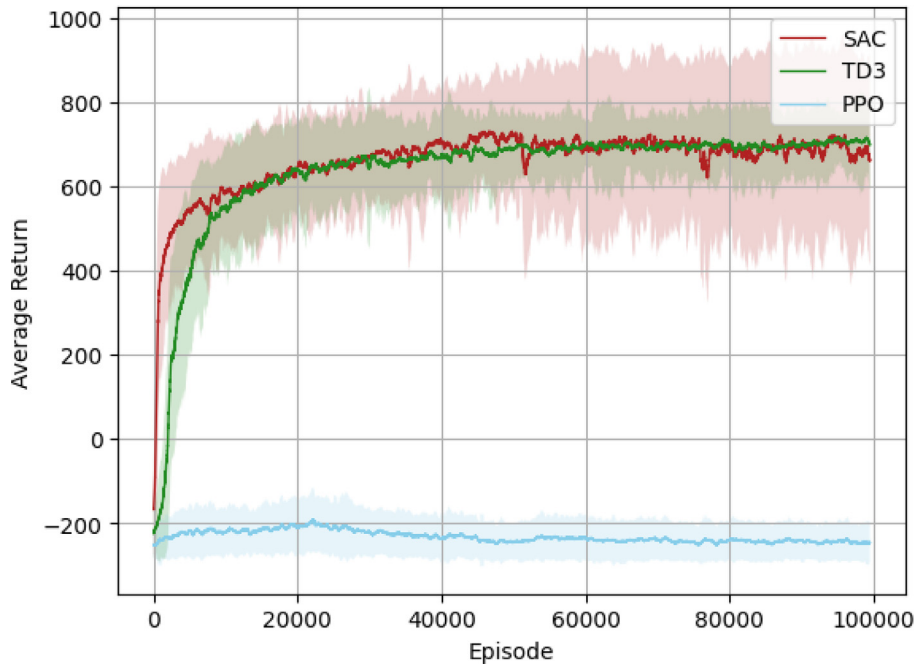


Fig. 4. Training curves on attitude control for spacecraft using CMGs.

return to their initial angles when the phase ends. Considering this aspect, the second phase also employs a momentum recovery steering law [Leeghim et al. \(2020\)](#).

$$\dot{\delta} = A^+ \tau_c + \eta(A^+ A - I_{4 \times 4}) \mathbf{g} \quad (26)$$

$$A^+ = A^T (A A^T)^{-1}$$

where η is a positive constant and the gradient vector $\mathbf{g} = (\delta - \delta_0)/\Delta t$ forces the gimbal angles to approach the initial gimbal angles δ_0 . Note that the second term of Eq. 26 is related to the null motion, which does not generate the torque. The conceptual flowchart of the DRL-based attitude control system is presented in Fig. 3.

Fig. 4 displays the training curves resulting from training with different deep reinforcement learning algorithms (SAC, TD3, PPO) according to the previously described training scenario. The solid lines represent the moving average return, and shaded regions represent the standard deviation with the moving window set to 1000.

As the training curves show, SAC and TD3 converge to high return values, indicating successful learning. In contrast, PPO's convergence to significantly lower return values suggests a poorer learning performance for this task. This is likely due to PPO's lower sample efficiency and more conservative exploration than the other two algorithms. Therefore, using PPO might require appropriate hyperparameter tuning.

SAC and TD3 stabilize at return values of about 700. Despite SAC exhibiting slight fluctuation in returns due to its unique exploration–exploitation trade-off, it rises faster than TD3 and converges to similar return values. This suggests that SAC has better sample efficiency and can effectively learn the attitude control policy. Given the nature of this study, which aims to achieve a desired attitude by steering four SGCMGs, high sample efficiency, and exploration performance are crucial, making SAC the most suitable algorithm.

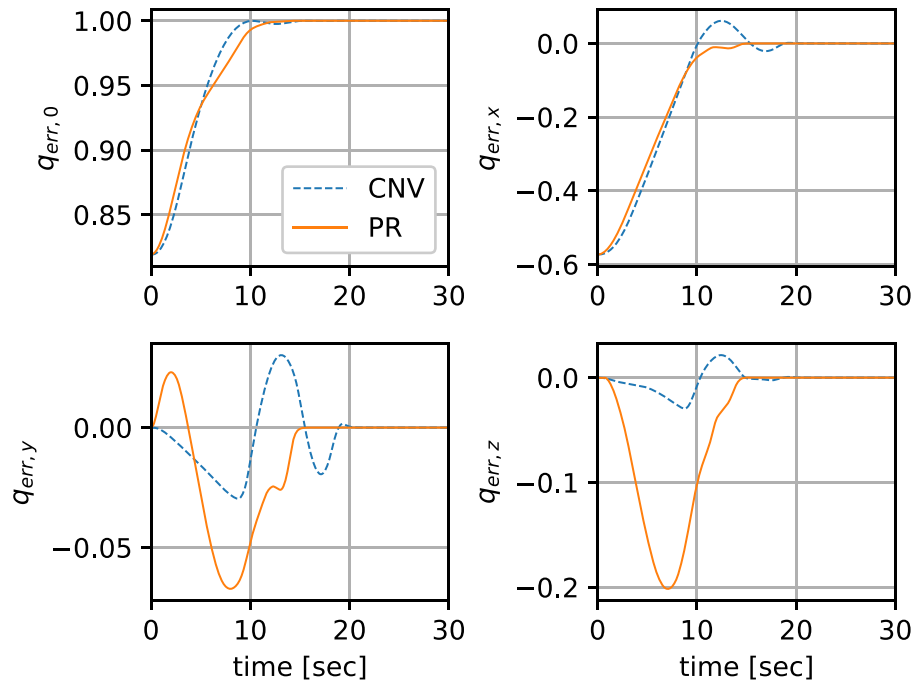


Fig. 5. Time history of the attitude error quaternions for the single-axis attitude control problem.

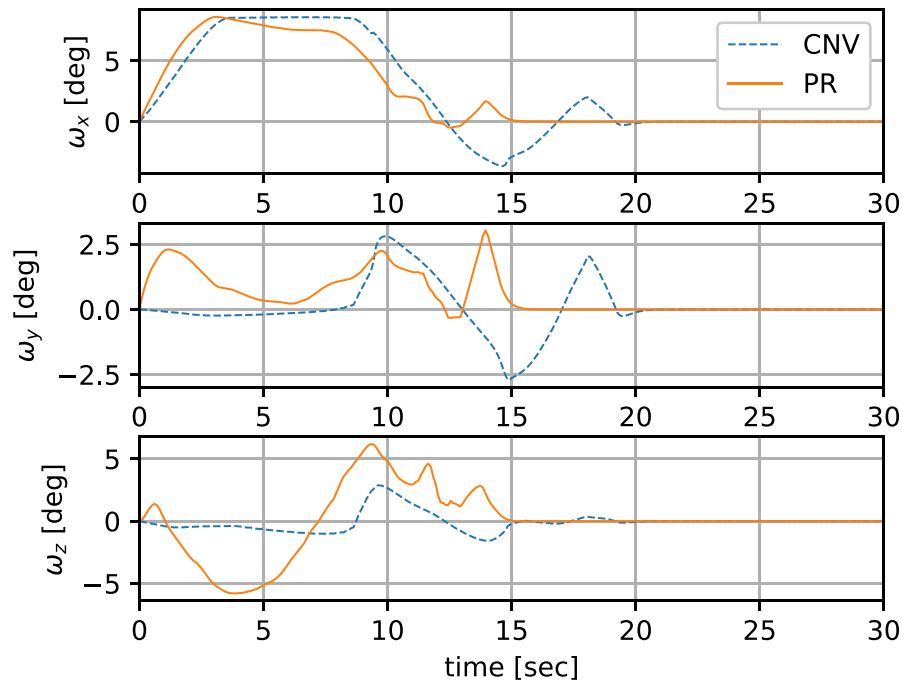


Fig. 6. Time history of the angular rates for the single-axis attitude control problem.

5. Simulation results

This section explores the effectiveness of the DRL-based attitude control system through simulation results. The simulations consider three cases: single-axis attitude con-

trol, three-axis attitude control, and multi-target tracking scenario. The proposed approach is compared and analyzed against the conventional approach composed of the quaternion-error feedback control law in Eq. 25 with the variable limiter and the generalized singularity robustness

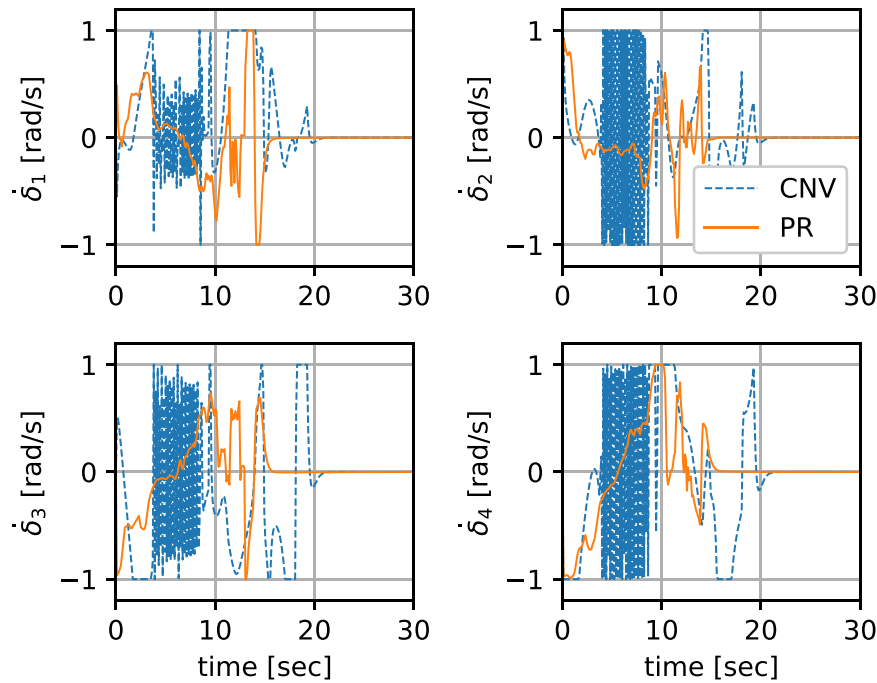


Fig. 7. Time history of the gimbal rates for the single-axis attitude control problem.

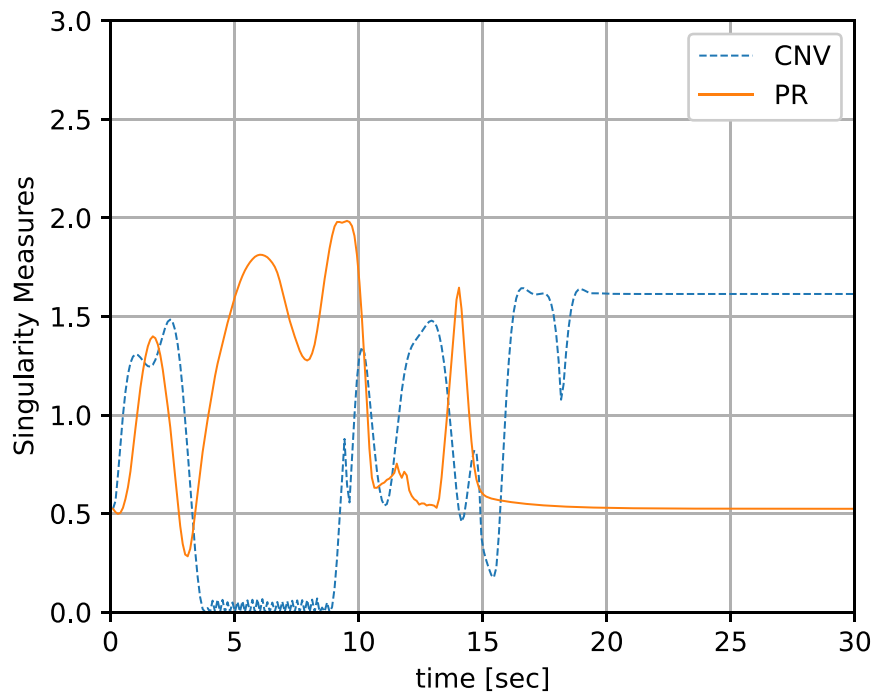


Fig. 8. Time history of the singularity measures for the single-axis attitude control problem.

inverse (GSRI) steering law. Note that the control law used in the conventional approach is identical to the one used in the proposed approach's second phase. In the subsequent simulation results, PR indicates the proposed DRL-based attitude control system, and CNV indicates the conventional approach used for comparison.

The GSRI steering law, used with the control law in Eq. 25, is widely employed as a simple yet effective method for passing through and escaping internal singularities [Wie et al. \(2002\)](#), [Bailey et al. \(2000\)](#). The proposed steering law is mainly intended for general maneuvers except pointing or tracking, fully utilizing the available momentum

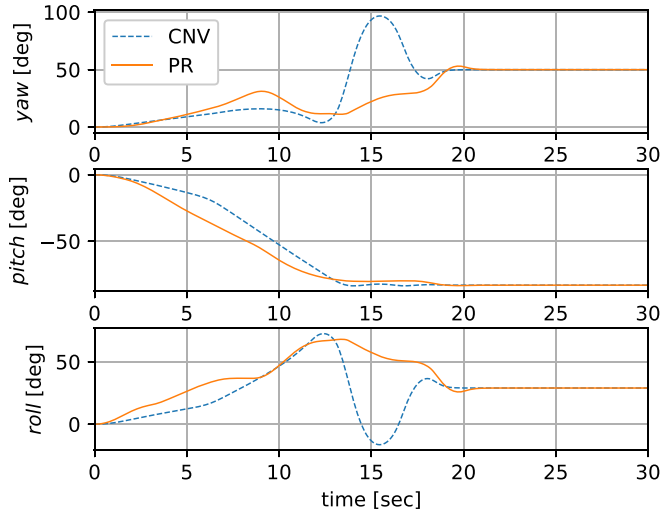


Fig. 9. The time histories of attitude angles for the three-axis attitude control problem.

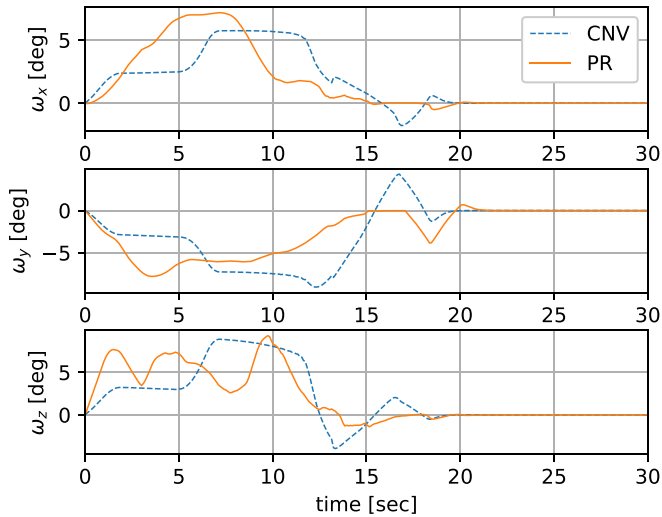


Fig. 10. The time histories of angular rates for the three-axis attitude control problem.

space of the CMG system even though singularities are present. The GSRI steering law is as follows.

$$\dot{\delta} = A^T [AA^T + \lambda E]^{-1} \tau_c \quad (27)$$

$$\lambda = 0.01 \exp [-10 \det (AA^T)]$$

$$E = \begin{bmatrix} 1 & \epsilon_3 & \epsilon_2 \\ \epsilon_3 & 1 & \epsilon_1 \\ \epsilon_2 & \epsilon_1 & 1 \end{bmatrix}$$

where the off-diagonal elements $\epsilon_i = 0.01 \sin (0.5\pi t + \phi_i)$ with $\phi_2 = \pi/2$, and $\phi_3 = \pi$.

5.1. Single-Axis Attitude Control

The first simulation considers a single-axis attitude control problem for a rigid spacecraft. The parameters of the spacecraft are assumed to be the same as those used in

training the agent with the DRL-based attitude control system, as listed in Table 2. For the control law of both the conventional and proposed approaches, the maximum control acceleration a was chosen as U/J , and the control torque command has a saturation limit of $3000Nm$, with the maximum slew rate, ω_{max} , assumed to be 30 degrees. Assuming $\omega_n = 3rad/s$, $\zeta = 0.9$, and $T = 10s$, the controller gains can be determined as follows: $k = \omega_n^2 + 2\zeta\omega_n/T$ and $c = 2\zeta\omega_n + 1/T$.

Fig. 5 and 6 show the simulation results of two approaches for a yaw attitude command of 70 degrees. Both approaches achieve the desired attitude when examining the attitude error quaternions and angular rates results. However, looking at the gimbal rates results, Fig. 7, when using the conventional approach, it can be seen that chattering starts to occur in 4 s. This phenomenon appears when the CMG system falls into a singularity, and it is essential to note that gimbal rates cannot be calculated when operating an actual CMG system in such a condition. To explore whether the CMG system has fallen into a singularity, the singularity measures, $m = \det(AA^T)$, is used (when the CMG system is in a singularity, m becomes 0). The graph for the singularity measures also approaches a value of 0 around 4 s, indicating that the CMG system has fallen into a singularity. In contrast, the DRL-based attitude control system achieves the desired attitude without the chattering of gimbal rates associated with singularities. These results imply that the proposed approach potentially circumvents the singularity problem in the CMG system. (see Fig. 8).

5.2. Three-axis attitude control

This section extends the single-axis attitude control problem to a three-axis attitude control problem, describing a typical spacecraft operation. The attitude commands are arbitrarily selected as $[50, -83, 29]$ degrees, assuming large-angle maneuvers.

The time histories of attitude angles and angular rates for both the conventional and proposed approaches are displayed in Fig. 9 and 10. Both approaches achieve the desired attitude in a similar amount of time. In Fig. 11, however, the conventional approach produces extreme chattering around 2 and 7 s, with the singularity measures approaching zero in Fig. 12. The DRL-based attitude control system produces no chattering at the gimbal rates despite the singularity measures approaching zero around 5 and 10 s.

5.3. Multi-target attitude control

This section presents a simulation study for multi-target attitude control, suitable for realistic missions. Each attitude command is generated every 30 s resulting in 8 attitude reorientations. Table 3 summarizes the randomly selected attitudes for this simulation.

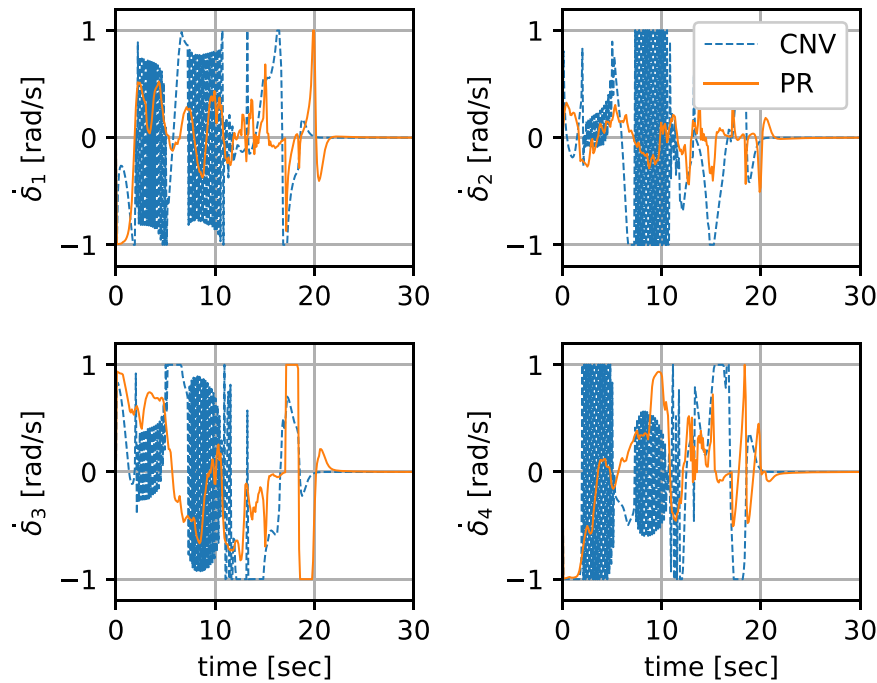


Fig. 11. The time histories of gimbal rates for the three-axis attitude control problem.

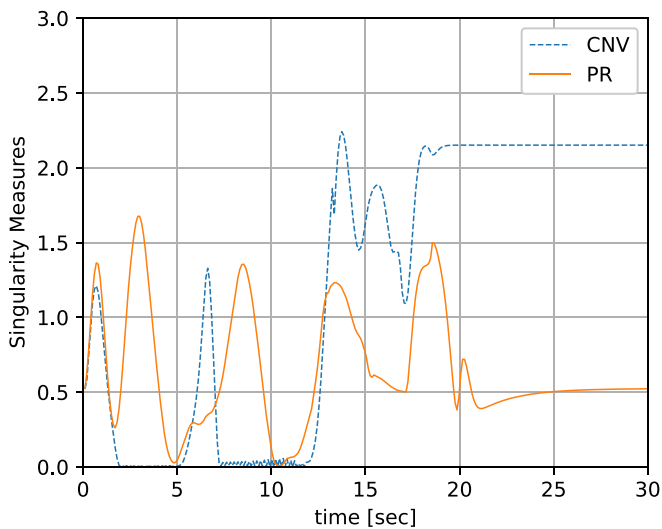


Fig. 12. The time histories of singularity measures for the three-axis attitude control problem.

Figs. 13 and 14 show the time histories of the attitude angles for the multi-target scenario. The DRL-based attitude control system (PR) and the conventional approach (CNV) perform well for the eight attitude commands. Fig. 15 shows the time history of the gimbal angles of the CMG system steered by each approach. Since the proposed approach includes momentum recovery driving, we can see that after achieving each command, the CMG system's gimbal angles return to the initial set angles. On the other hand, the conventional approach does not include this consideration, resulting in a significant deviation from the ini-

Table 3

Desired Euler angles for multi-target scenario.

Target Number	Desired Attitude [yaw, pitch, roll]
1	[14, -1, -11] (deg)
2	[-67, -50, 5] (deg)
3	[3, 57, 73] (deg)
4	[-4, -52, -18] (deg)
5	[40, 5, -85] (deg)
6	[-84, -45, -39] (deg)
7	[25, 42, 12] (deg)
8	[-51, 61, -61] (deg)

tialized gimbal angles after successive attitude control. These results have important implications for the CMG system, as the initial position of the gimbal affects its performance. Fig. 16 and 17 also show that similar to the results for the three-axis attitude control problem, the traditional method results in extreme chattering due to the singularity problem.

5.4. Robustness analysis

The final aspect to examine is the robustness of the proposed approach. The DRL-based attitude control system defines the agent's state using attitude error, angular velocity, and gimbal angles. This means attitude control can be performed without relearning, even if the model parameters change. To demonstrate this, we use different model parameters from those used in the training and consider the external disturbances. We also added white Gaussian noise to assume measurement errors in the angular momentum, angular velocity, and gimbal angles of the

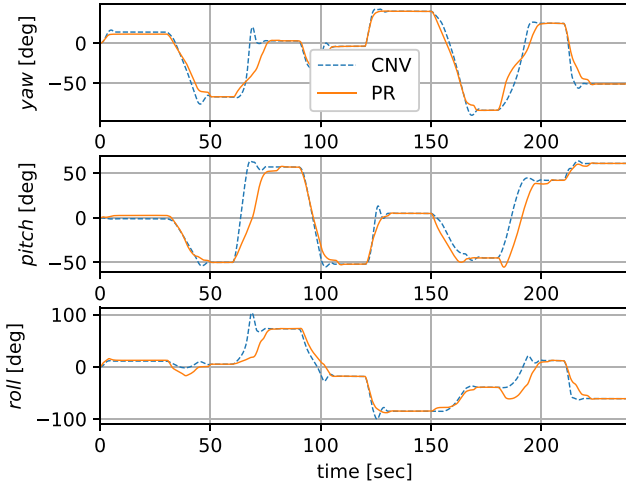


Fig. 13. The time histories of the attitude angles for multi-target scenario.

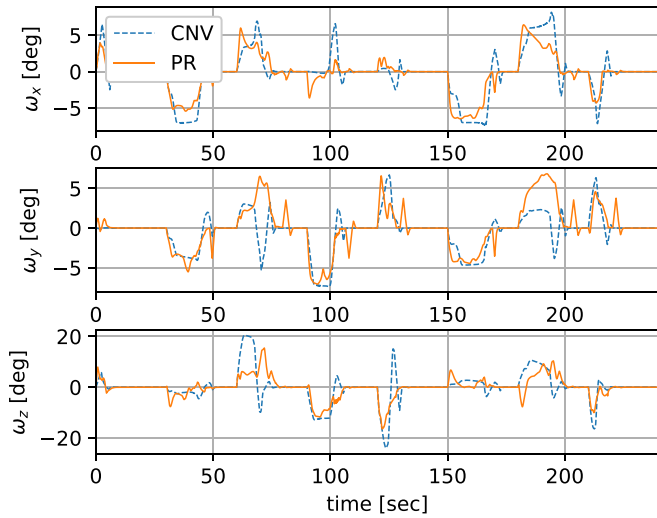


Fig. 14. The time histories of the angular rates for the multi-target scenario.

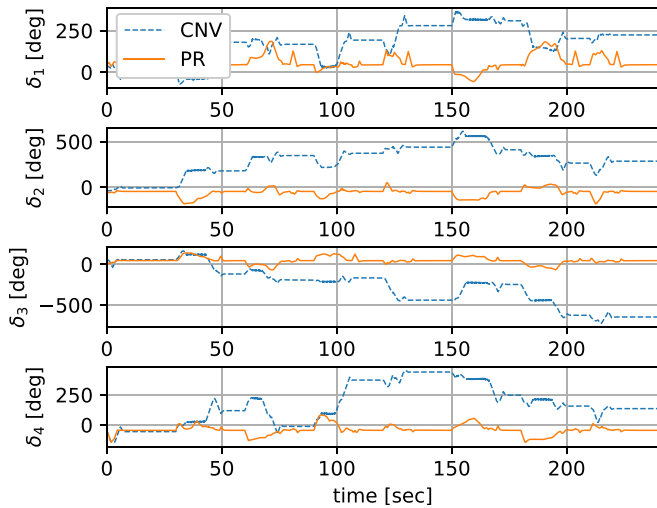


Fig. 15. The time histories of the gimbal angles for the multi-target scenario.

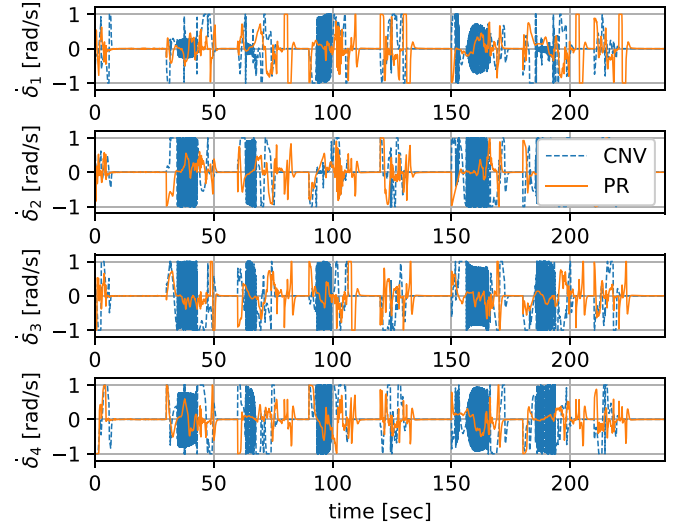


Fig. 16. The time histories of the gimbal angles for multi-target scenario.

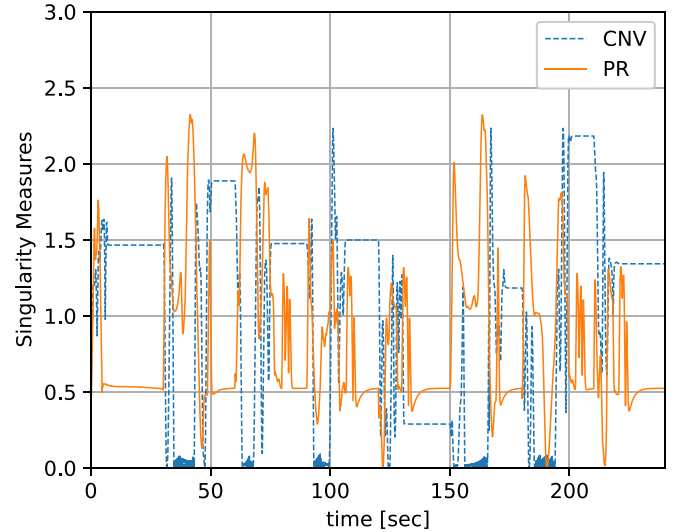


Fig. 17. The time histories of the singularity measures for the multi-target scenario.

SGCMGs. Furthermore, the configuration of the SGCMGs cluster is changed from a pyramid-type to a roof-type for the simulations. Table 4 summarizes the changed model parameters and noise, and the external disturbance is chosen as

$$\tau_{ext} = \begin{bmatrix} 5 \sin(0.01t + 0.25\pi) \\ -5 \sin(0.01t + 0.5\pi) \\ 10 \sin(0.01t + 0.7\pi) \end{bmatrix} \quad (28)$$

The Jacobian matrix of the roof-type SGCMG cluster is as follows:

$$A_{roof} = \begin{bmatrix} \sin \delta_1 & \sin \delta_2 & -\sin \delta_3 & -\sin \delta_4 \\ -\cos \beta_r \cos \delta_1 & -\cos \beta_r \cos \delta_2 & \cos \beta_r \cos \delta_3 & \cos \beta_r \cos \delta_4 \\ \sin \beta_r \cos \delta_1 & \sin \beta_r \cos \delta_2 & \sin \beta_r \cos \delta_3 & \sin \beta_r \cos \delta_4 \end{bmatrix} \quad (29)$$

Table 4

Changed model parameters and noise levels to demonstrate the robustness of the proposed approach.

Parameter	Value	Unit
Moment of inertia, J	$diag(22470, 21105, 5250)$	kg·m ²
Angular momentum of SGCMG, H_{cmg}	800	N·m
Std. of angular momentum, σ_h	1.0	N·m
Std. of angular velocity noise, σ_ω	0.005	rad/s
Std. of gimbal angle noise, σ_δ	0.01	rad

where β_r denotes the skew-angle of the roof-type configuration and is set to 45 degree.

The simulation results, presented in Fig. 18–21, highlight the adaptability and versatility of our proposed approach. The blue dashed line represents the results without disturbances and noise, and the orange solid line repre-

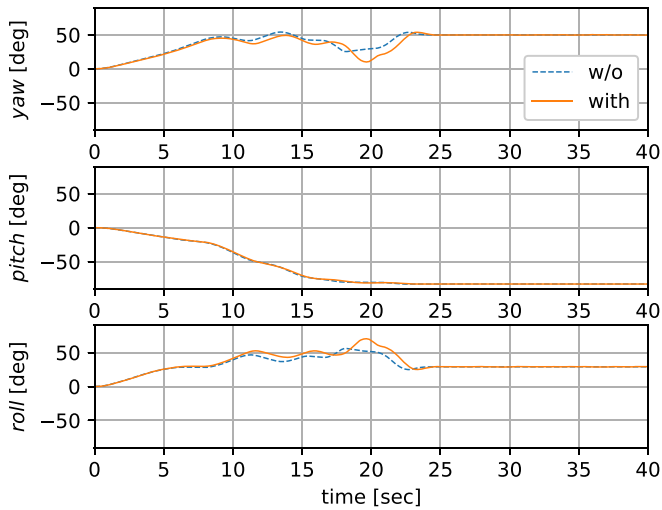


Fig. 18. Time histories of Euler angles for pyramid types and scenario taking into account model uncertainties, external disturbance, and noise.

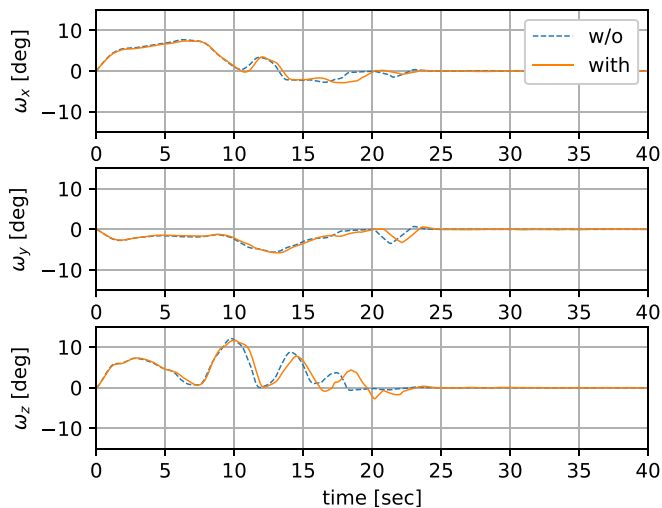


Fig. 19. Time histories of angular rates for pyramid types and scenario taking into account model uncertainties, external disturbance, and noise.

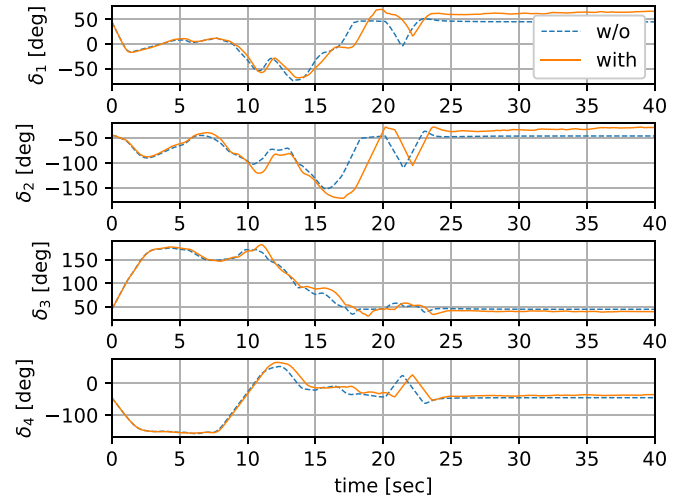


Fig. 20. Time histories of gimbal angles for pyramid types and scenario taking into account model uncertainties, external disturbance, and noise.

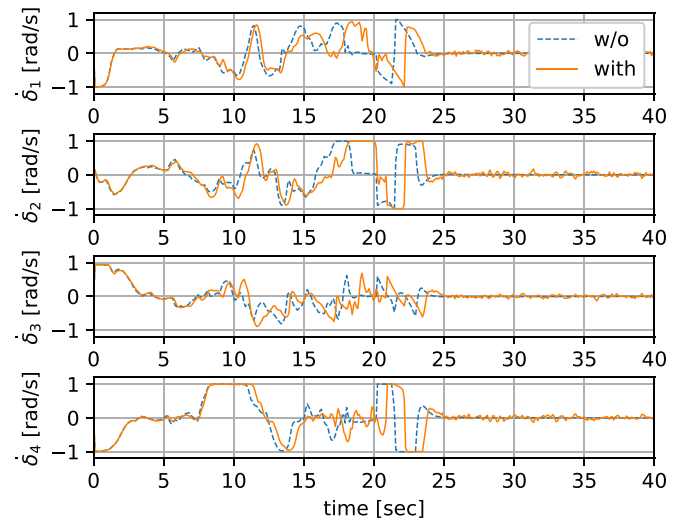


Fig. 21. Time histories of gimbal rates for pyramid types and scenario taking into account model uncertainties, external disturbance, and noise.

sents these factors considered. Although there is a slight delay due to disturbances and noise, the proposed approach achieves the desired attitude. These results show the effectiveness of the proposed approach to different model parameters and various type configurations, demonstrating its versatility and reassuring its potential in diverse scenarios.

However, starting at about 23 s in Fig. 20, the gimbal angle shows that it is slightly biased and does not recover the initial gimbal angle. This section, which begins the second phase of precise postural control and momentum vector recovery, uses a conventional approach. The results mean that the second phase still requires considering uncertainties, perturbations, and noise, unlike the first phase, which uses DRL-based attitude control.

6. Conclusion

The attitude control problem of a rigid spacecraft using the CMG system is a well-known research topic. While various steering laws have been studied to address the singularity problem of the CMG system, a steering law that can fully utilize their performance while avoiding/escaping singularities has yet to be developed. In this study, we propose a DRL-based attitude control system to overcome the singularity problem of single-gimbal control moment gyros (SGCMGs) and achieve the desired attitude.

We defined the observation of the agent as the attitude error quaternions, the angular rates of the spacecraft, and the gimbal angles of the SGCMGs. The agent's action is defined as the gimbal rates of the SGCMGs. In addition, the reward function is designed considering the characteristics of the spacecraft attitude control problem using SGCMGs. DRL techniques are used to solve the Markov decision process (MDP) defined in this way, and among them, SAC algorithms specialized in control tasks are used. Since the SGCMGs are steered based on sampling data through the DRL techniques, they can bypass the singularity problem.

However, achieving high precision in attitude control and maintaining a steady state using only DRL techniques takes much work. Therefore, we divide the attitude control system into two phases. The first phase is based on the DRL, and the second is based on a conventional approach. The conventional approach comprises the quaternion error feedback control and the momentum recovery steering law. Once the DRL technique performs the attitude control to a certain level, the conventional approach performs precise control.

Finally, we explored the validity of the proposed approach in various simulations. Simulation results have shown that the desired attitude is achieved by steering SGCMGs well without any singularity problem, and it is noteworthy that attitude control is performed similarly to the conventional approach. In practical cases, the performance and versatility of the proposed approach are further highlighted by the fact that the conventional approach cannot achieve the same level of performance as the simulation results due to issues such as model uncertainty, disturbances, noise, and the singularity problem of CMG systems.

Previous studies applying machine learning techniques have often used them as auxiliary means alongside other methodologies. In contrast, this study demonstrated that reinforcement learning can directly steer actuators to achieve attitude control in an end-to-end manner. This suggests that data-driven control approaches like reinforcement learning have recently gained significant attention and can be crucial in addressing the singularity problem. These results can provide valuable insights for future research in attitude control using the CMG system.

We presented a novel approach to overcome the singularity problem of SGCMGs, but some limitations still need

to be addressed in future research. The proposed approach defines the satellite's attitude error, angular velocity, and SGCMG gimbal angle as the agent's state. Therefore, attitude control can be performed well without retraining, even if the model parameters change. However, new training is required if the state space or action space is changed, for example, when using a triple-parallel configuration or a variable-speed CMGs cluster. The second limitation is the inherent reliability issue of reinforcement learning. Since control is performed through deep neural networks, theoretical analysis such as stability assessment, which is used in control theory, is difficult. We indirectly addressed these issues by combining reinforcement learning with traditional control methods, and explainable AI, which is actively researched in machine learning, can also help analyze and explain the reliability of reinforcement learning-based controllers. However, the reliability of reinforcement learning-based attitude controllers remains an issue that needs to be resolved.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the 'Space Pioneer Program' grant funded by the Ministry of Science and ICT, Republic of Korea. (2021M1A3B9094394)

References

- Bailey, D.A., Heiberg, C.J., Wie, B., 2000. Continuous attitude control that avoids cmg array singularities. US Patent 6,131,056.
- Bedrossian, N.S., Paradise, J., Bergmann, E.V., et al., 1990. Steering law design for redundant single-gimbal control moment gyroscopes. *J. Guid., Control, Dynam.* 13 (6), 1083–1089.
- Bunryo, Y., Satoh, S., Shoji, Y., et al., 2021. Feedback attitude control of spacecraft using two single gimbal control moment gyros. *Adv. Space Res.* 68 (7), 2713–2726.
- Cornick, D., 1979. Singularity avoidance control laws for single gimbal control moment gyros. In: *Guidance and Control Conference* (p. 1698).
- Davis, B., 1972. A Comparison of CMG Steering Laws for High Energy Astronomy Observatories (HEAOS). Technical Report Marshall Space Flight Center.
- François-Lavet, V., Henderson, P., Islam, R., et al., 2018. An introduction to deep reinforcement learning. *Found. Trends Mach. Learn.* 11 (3–4), 219–354.
- Fujimoto, S., Hoof, H., Meger, D., 2018. Addressing function approximation error in actor-critic methods. In: *International Conference on Machine Learning*. PMLR, pp. 1587–1596.
- Haarnoja, T., Zhou, A., Abbeel, P., et al., 2018. Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor. In: *International Conference on Machine Learning*. PMLR, pp. 1861–1870.
- Hua, B., Ni, R., Zheng, M., et al., 2023. Cooperative game theory-based steering law design of a cmg system. *J. Syst. Eng. Electron.* 34 (1), 185–196.

- Kuhns, M.D., Rodriguez, A.A., 1995. A preferred trajectory tracking steering law for spacecraft with redundant cmgs. In: Proceedings of 1995 American Control Conference-ACC'95 (pp. 3111–3115). IEEE volume 5.
- Lee, J.-S., Bang, H.-C., Lee, H.-J., 2005. Singularity avoidance by game theory for control moment gyros. In: AIAA Guidance, Navigation, and Control Conference and Exhibit (p. 5946).
- Leehim, H., Bang, H.-C., Park, J.-O., 2009. Singularity avoidance of control moment gyros by one-step ahead singularity index. *Acta Astronaut.* 64, 935–945.
- Leehim, H., Jin, J., Mok, S.-H., 2018. Feasible angular momentum of spacecraft installed with control moment gyros. *Adv. Space Res.* 61 (1), 466–477.
- Leehim, H., Kim, D., 2015. Adaptive neural control of spacecraft using control moment gyros. *Adv. Space Res.* 55 (5), 1382–1393.
- Leehim, H., Lee, C.-Y., Jin, J.-H., et al., 2020. A singularity-free steering law of roof array of control moment gyros for agile spacecraft maneuver. *Int. J. Control Autom. Syst.* 18, 1679–1690.
- Liang, T., Xu, S.-J., 2005. Geometric analysis of singularity for single-gimbal control moment gyro systems. *Chin. J. Aeronaut.* 18 (4), 295–303.
- Lillicrap, T.P., Hunt, J.J., Pritzel, A. et al. (2015). Continuous control with deep reinforcement learning. arXiv preprint arXiv:1509.02971.
- Mnih, V., Kavukcuoglu, K., Silver, D. et al. (2013). Playing atari with deep reinforcement learning. arXiv preprint arXiv:1312.5602.
- Montoya-Cháirez, J., Rossomando, F.G., Carelli, R., et al., 2021. Adaptive rbf neural network-based control of an underactuated control moment gyroscope. *Neural Comput. Appl.* 33 (12), 6805–6818.
- Mony, A., Paranjape, A.A., 2022. Singularity avoidance in spacecraft with single-gimbal cmgs using rrt-based steering laws. In: AIAA Scitech 2022 Forum (p. 0864).
- Papakonstantinou, C., Daramouskas, I., Lappas, V., et al., 2022. A machine learning approach for global steering control moment gyroscope clusters. *Aerospace* 9 (3), 164.
- Paradiso, J.A., 1992. Global steering of single gimbaled control moment gyroscopes using a directed search. *J. Guid., Control, Dynam.* 15 (5), 1236–1244.
- Petersen, C.D., Leve, F., Kolmanovsky, I., 2017. Model predictive control of an underactuated spacecraft with two reaction wheels. *J. Guid., Control, Dynam.* 40 (2), 320–332.
- Schulman, J., Levine, S., Abbeel, P., et al., 2015. Trust region policy optimization. In: International Conference on Machine Learning. PMLR, pp. 1889–1897.
- Schulman, J., Wolski, F., Dhariwal, P. et al., 2017. Proximal policy optimization algorithms. arXiv preprint arXiv:1707.06347.
- Seo, H.-H., Bang, H.-C., Cheon, Y.-J., 2019. Steering law of control moment gyros using artificial potential function approach. *Acta Astronaut.* 157, 374–389.
- Sutherland, R., Kolmanovsky, I., Girard, A.R., 2018. Attitude control of a 2u cubesat by magnetic and air drag torques. *IEEE Trans. Control Syst. Technol.* 27 (3), 1047–1059.
- Sutton, R.S., Barto, A.G., 2018. Reinforcement Learning: An Introduction. MIT press.
- Takada, K., Kojima, H., Matsuda, N., 2010. Control moment gyro singularity-avoidance steering control based on singular-surface cost function. *J. Guid., Control, Dynam.* 33 (5), 1442–1450.
- Wang, P., & Shtessel, Y.B. (1998). Satellite attitude control using only magnetorquers. In Proceedings of the 13th Southeastern Symposium on System Theory (pp. 500–504).
- Wie, B., 2004. Singularity analysis and visualization for single-gimbal control moment gyro systems. *J. Guid., Control, Dynam.* 27 (2), 271–282.
- Wie, B., 2005. Singularity escape/avoidance steering logic for control moment gyro systems. *J. Guid., Control, Dynam.* 28 (5), 948–956.
- Wie, B., Bailey, D., Heiberg, C., 2001. Singularity robust steering logic for redundant single-gimbal control moment gyros. *J. Guid., Control, Dynam.* 24 (5), 865–872.
- Wie, B., Bailey, D., Heiberg, C., 2002. Rapid multitarget acquisition and pointing control of agile spacecraft. *J. Guid., Control, Dynam.* 25 (1), 96–104.
- Wie, B., Bailey, D.A., & Heiberg, C.J. (2000). Robust singularity avoidance in satellite attitude control. U.S. Patent 6,039,290.
- Yeh, F.-K., 2010. Sliding-mode adaptive attitude controller design for spacecrafts with thrusters. *IET Control Theory Appl.* 4 (7), 1254–1264.
- Yoon, H.-J., Tsiotras, P., 2004. Singularity analysis of variable speed control moment gyros. *J. Guid., Control, Dynam.* 27 (3), 374–386.
- Yuandong, L., Qinglei, H., Xiaodong, S., 2022. Neural network-based fault diagnosis for spacecraft with single-gimbal control moment gyros. *Chin. J. Aeronaut.* 35 (7), 261–273.
- Zhang, K., Wang, S., Wang, S., et al., 2023. Anomaly detection of control moment gyroscope based on working condition classification and transfer learning. *Appl. Sci.* 13 (7), 4259.
- Zhao, H., Liu, M., Sun, Y., et al., 2022. Fault diagnosis of control moment gyroscope based on a new cnn scheme using attention-enhanced convolutional block. *Sci. China Technol. Sci.* 65 (11), 2605–2616.
- Ziebart, B.D., 2010. Modeling Purposeful Adaptive Behavior with the Principle of Maximum Causal Entropy. Carnegie Mellon University.